

Project Report: Intelligent Travel Recommendation System — Canada

MSc Data Science Project

Subject: Hybrid Recommendation Engine for Canadian Tourism

1. Executive Summary

The **Intelligent Travel Recommendation System** is a sophisticated academic prototype designed to deliver personalized travel itineraries across Canada. By synthesizing **Content-Based Filtering (CBF)** and **Collaborative Filtering (CF)**, the system achieves a balanced approach to the "cold-start" problem and long-term user personalization. The engine is grounded in a synthetic dataset of 5,462 ratings modeled after Statistics Canada tourism structures.

2. System Architecture & Methodology

The system utilizes a multi-layered ML pipeline to process destination features and user preferences.

2.1 Feature Engineering

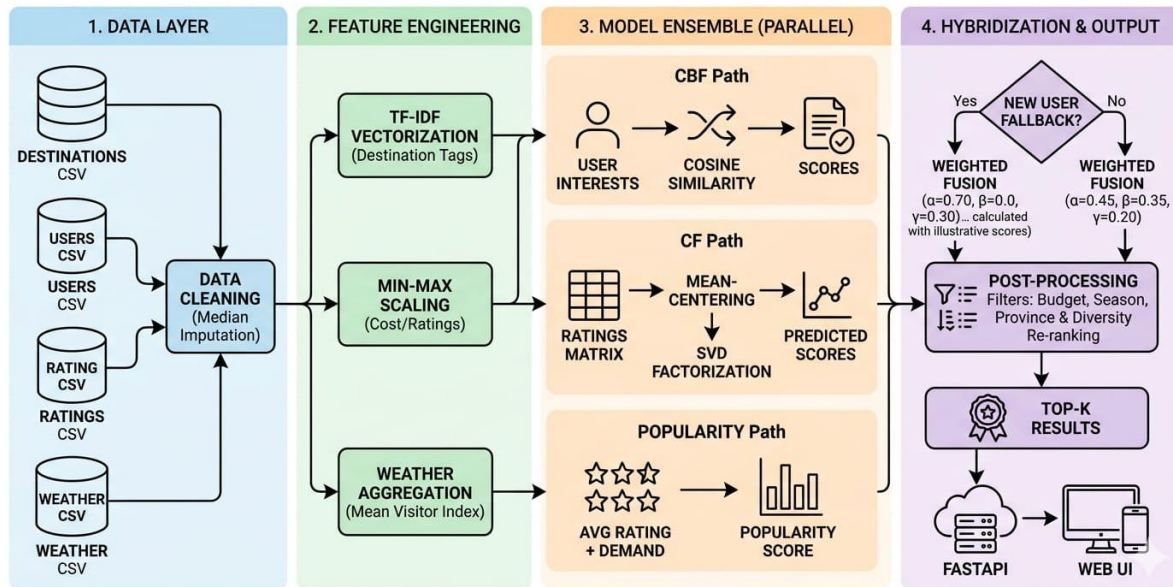
- **Textual Features:** TF-IDF vectorization is applied to destination "interest tags" (e.g., *nature*, *hiking*, *history*) to create a high-dimensional content space.
- **Numerical Features:** `avg_cost_day`, `avg_rating`, and a custom `visitor_index` are normalized using Min-Max scaling.
- **Temporal Intelligence:** A seasonal visitor index incorporates climate data to adjust recommendations based on monthly demand.

2.2 Recommendation Engine (Hybrid Ensemble)

The core scoring logic follows a weighted ensemble approach:

- **Formula:** $\text{Score} = 0.45 * \text{CBF} + 0.35 * \text{CF} + 0.20 * \text{Popularity}$.
- **Collaborative Filtering:** Employs **Truncated SVD** (Singular Value Decomposition) to identify latent factors in the user-destination interaction matrix.
- **Cold-Start Strategy:** For new users without history, the system automatically pivots to a **0.70 CBF + 0.30 Popularity** blend.

A Hybrid Travel Recommendation System Architecture



3. Dataset Characteristics

The project operates on a comprehensive synthetic ecosystem generated to mimic real-world tourism patterns:

- **Destinations:** 73 cataloged locations across all Canadian provinces.
- **Users:** 500 unique profiles with varying budgets and interest combos.
- **Interactions:** 5,462 ratings with an observed sparsity of ~85% in the interaction matrix.
- **Supplementary Data:** 250 records each for hotels, attractions, and dining options to provide full itinerary context.

4. Performance Evaluation

Offline evaluation was conducted using a sample of 40 users with a cutoff of **K=10**.

Metric	Score	Insight
Precision@10	0.216	The percentage of recommended items that are relevant to the user.
Recall@10	0.648	The system's ability to find all relevant items within the top 10.

Metric	Score	Insight
NDCG@10	0.467	Measures the quality of the ranking (higher is better).
RMSE	0.086	The average error in rating prediction.

Key Finding: Sensitivity analysis confirmed that an **alpha weight of 0.45** for Content-Based Filtering yields the optimal balance for NDCG and Precision across the synthetic dataset.

5. Technical Implementation

- **Backend:** FastAPI provides a robust REST interface for real-time recommendation delivery.
 - **Frontend:** An interactive HTML/CSS presentation serves as a demo interface for stakeholders.
 - **Deployment:** Containerized via Docker to ensure reproducibility across different environments.
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6. Conclusion & Future Scope

This prototype successfully demonstrates a production-grade recommendation pipeline. While currently utilizing synthetic data, the modular architecture allows for future integration of live travel APIs and sentiment analysis on real user reviews.